

report

April 25, 2025

```
[ ]: import socket
from flask import Flask, render_template, request, redirect, jsonify, session
import requests
import os

app = Flask(__name__)
app.secret_key = os.urandom(24)
```

```
[ ]: def get_local_ip():
    """
        IP-
        UDP-          DNS Google.
        '192.168.x.x'   '127.0.0.1'
    """
    try:
        s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
        s.connect(("8.8.8.8", 80))
        local_ip = s.getsockname()[0]
        s.close()
        return local_ip
    except Exception:
        return "127.0.0.1"

DEFAULT_ESP_IP = 'http://192.168.1.100'
```

```
def get_esp_ip():
    """
        IP ESP-          Flask.
        ,          DEFAULT_ESP_IP.
    """
    return session.get('esp_ip', DEFAULT_ESP_IP)
```

```
[ ]: @app.route('/')
def index():
    """
        :          control_panel          IP          .
    """
```

```
return render_template('index.html', current_ip=get_esp_ip())
```

```
[ ]: @app.route('/update_ip', methods=['POST'])
def update_ip():
    """
        POST-IP .
        , http:// https://

    """
    new_ip = request.form.get('ip')
    if new_ip and new_ip.startswith(('http://', 'https://')):
        session['esp_ip'] = new_ip
        return jsonify({'status': 'success', 'ip': new_ip})
    return jsonify({'status': 'error'}), 400
```

```
@app.route('/power/<state>')
def power_control(state):
    """
        LED- .
        state=='on' /on, /off.
        '/' HTTP 302.

    """
    esp_ip = get_esp_ip()
    if state == 'on':
        requests.get(f'{esp_ip}/on')
    else:
        requests.get(f'{esp_ip}/off')
    return redirect('/')
```

```
@app.route('/color')
def color():
    """
        : r, g, b query-string.

    """
    esp_ip = get_esp_ip()
    r = request.args.get('r')
    g = request.args.get('g')
    b = request.args.get('b')
    if all([r, g, b]):
        requests.get(f'{esp_ip}/color?r={r}&g={g}&b={b}')
    return redirect('/')
```

```
@app.route('/mode', methods=['POST'])
```

```

def set_mode():
    """
        :param mode: mode of the device.
        :param esp_ip: IP address of the ESP.
    """
    esp_ip = get_esp_ip()
    mode = request.form.get('mode', '0')
    requests.get(f'{esp_ip}/mode?m={mode}')
    return ('', 200)

```

```

[ ]: if __name__ == '__main__':
    """
        Flask-IP 5050.
    """
    app.run(host=get_local_ip(), port=5050)

```